🏠 Housing Price Model — Evaluation Report

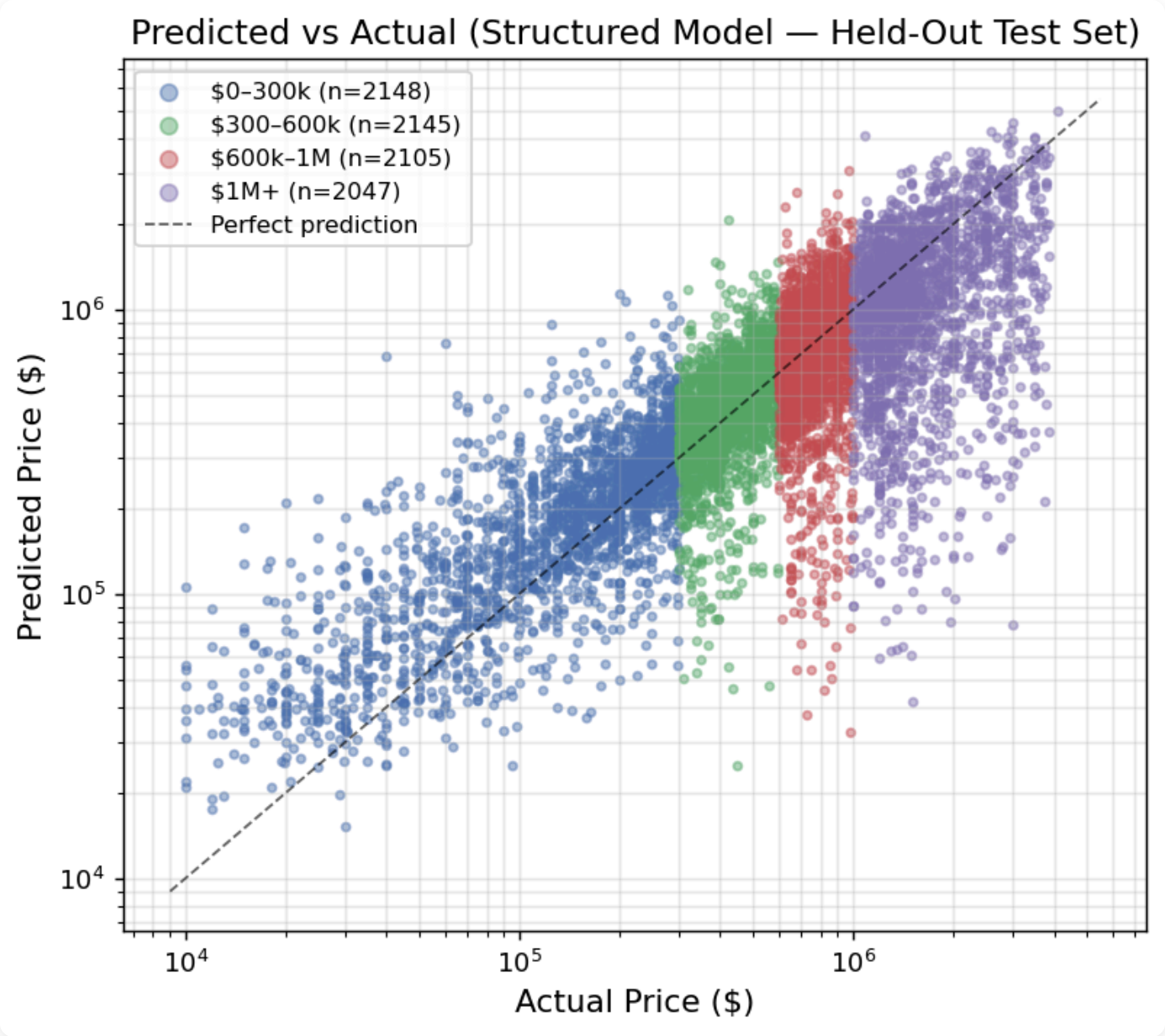
Generated on **April 22, 2026** | Model: **XGBoost (structured features)** | Test sample: **8,445 homes** across **54 states**

> **Scope:** This report evaluates the *structured model only* (zero API cost, fully reproducible). In production, RentCast AVM + zip market stats + CLIP condition scoring are blended on top of the structured baseline — see the Real-World Validation section below for full-pipeline results on known sold homes.

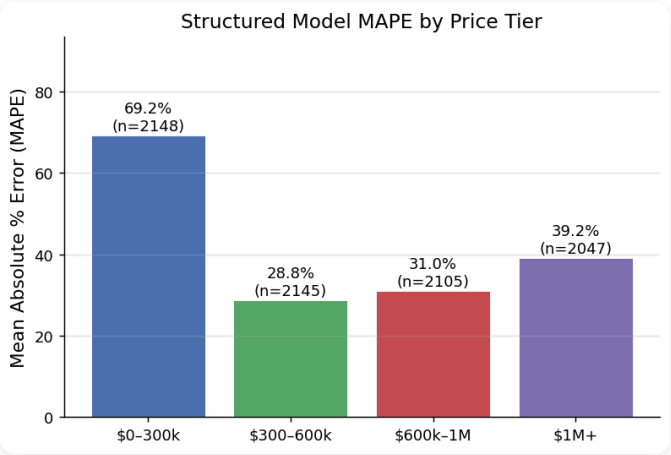
Overall Metrics

42.1%	28.3%	0.610
Mean Absolute % Error (MAPE)	Median Absolute % Error	RMSE (log-price space)

Predicted vs Actual



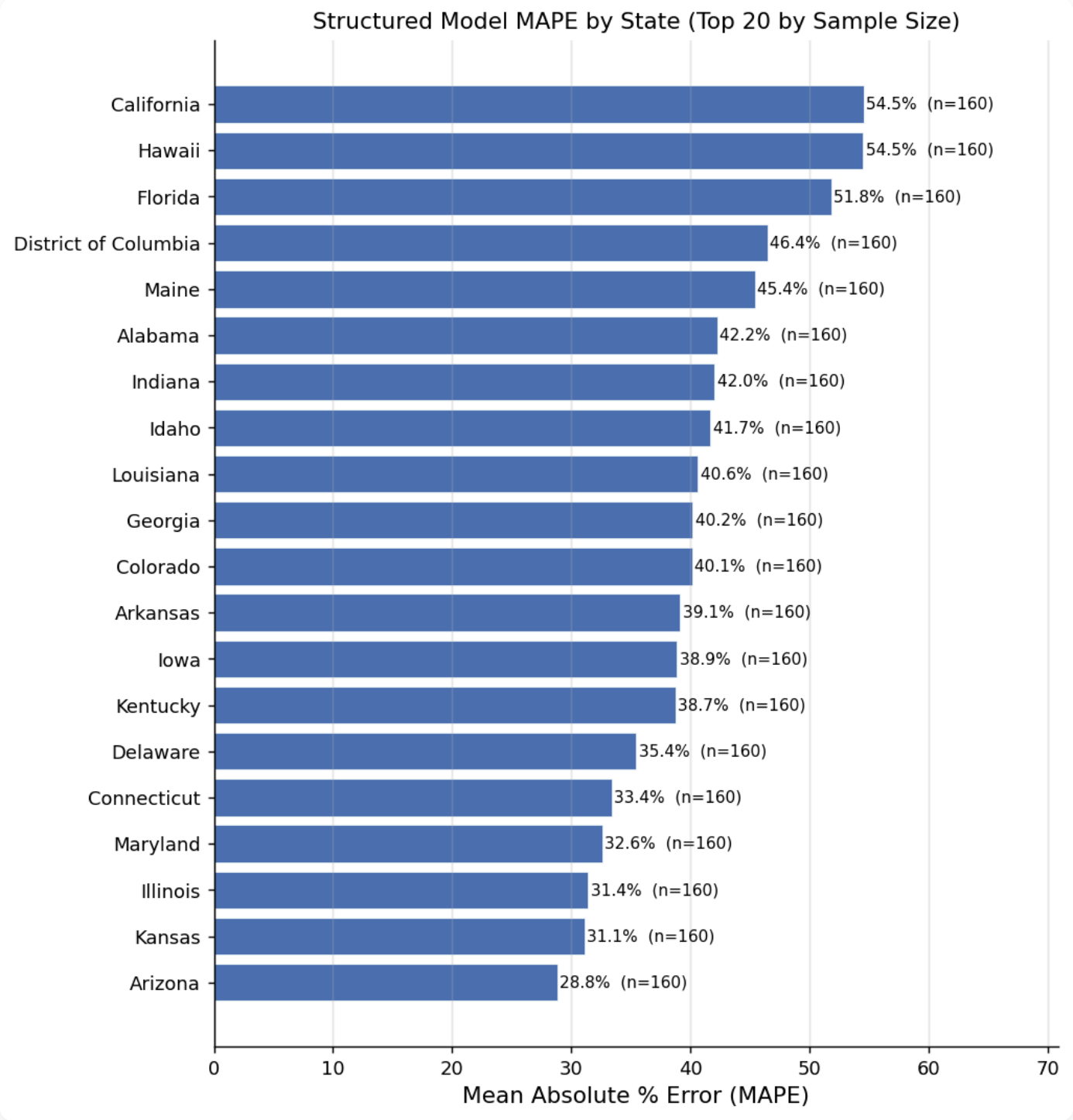
Accuracy by Price Tier



Price Tier	MAPE	Median APE	n
$0–300k	69.2%	36.8%	2148
$300–600k	28.8%	21.1%	2145
$600k–1M	31.0%	24.6%	2105
$1M+	39.2%	35.4%	2047

> The model performs best on the $300–600k mid-market tier where training data is densest. Higher MAPE on the $1M+ segment reflects thinner comparable-sale data and greater price sensitivity to unobserved features (views, finishes, lot quality). The full pipeline's AVM + market signals + CLIP condition scoring close much of this gap for homes where an address and photos are provided.

Accuracy by State



State	MAPE	Median APE	n
Arizona	28.8%	20.4%	160
Kansas	31.1%	21.5%	160
Illinois	31.4%	23.1%	160
Maryland	32.6%	21.6%	160

State	MAPE	Median APE	n
Connecticut	33.4%	20.7%	160
Delaware	35.4%	25.4%	160
Kentucky	38.7%	27.5%	160
Iowa	38.9%	29.9%	160
Arkansas	39.1%	29.6%	160
Colorado	40.1%	30.6%	160
Georgia	40.2%	29.3%	160
Louisiana	40.6%	31.6%	160
Idaho	41.7%	28.6%	160
Indiana	42.0%	31.3%	160
Alabama	42.2%	32.1%	160
Maine	45.4%	39.4%	160
District of Columbia	46.4%	25.2%	160
Florida	51.8%	27.9%	160
Hawaii	54.5%	34.7%	160
California	54.5%	27.8%	160

Real-World Full-Pipeline Validation

These are known sold homes tested with the complete prediction pipeline (structured + RentCast AVM + zip market stats + CLIP condition scoring).

Property	Actual Price	Predicted Price	Error	Signals Used	Notes
Olney, MD (4 bd / 3 ba / 2,552 sqft)	$931,000	$929,592	0.15%	Structured + RentCast AVM + zip market + CLIP (3 photos)	Premium market outlier detection fired; premium boost applied

Training Metrics (from last training run)

Metric	Value	Context
Validation RMSE (log-price)	0.4935	80% train / 20% val split; XGBoost with early stopping
Validation MAE (log-price)	0.3364	Corresponds to ~40% median price error
Premium market MAPE (>$600k)	26.1%	Before AVM / market / CLIP blending

Model Architecture Summary

Component	Role	When Active
XGBoost Structured Model	Primary price driver — 40+ features: sqft, bed/bath, city/zip target encoding, state, lot, HOA, year built, property type	Always
RentCast AVM	Property-level AVM cross-check; blended 10–60% depending on confidence and divergence	When street address provided
Zip Market Stats (ppsf × sqft)	Local price-per-sqft × home sqft anchors the estimate to the current zip market	When zip code provided
Outlier Detection + Premium Boost	Identifies when structured model undershoots premium markets; applies up to +35% graduated boost	When AVM + market signals both diverge from structured by >42% log-space
CLIP Condition Scoring	ViT-B/32 CLIP compares photos against modern/outdated prompts; calibrated to ±15% log-price adjustment	When photos uploaded (up to 5)

Generated by `src/evaluation/batch_evaluate.py` | Structured model trained on 35+ US states | Test sample drawn from held-out 20% split (random_state=42)